EXPERIMENT 8

Configuring User Management and Role-Based Access Control (RBAC)

Gym Membership Management System – Administrator and Trainer roles

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Application	Gym_Membership_Management_System (Zoho Creator)	**Department**	B.Tech CSBS

1. Aim

To plan and describe user management and role-based access control (RBAC) for the gym application: the Administrator and Trainer users, the permission sets, a role hierarchy, and a data-sharing rule, so that each person sees and changes only what the job requires.

Objectives

- Review the gym application and its dashboard.
- Study the permission sets (Read and Write) under User Permissions.
- Design the role hierarchy CEO → Gym Administrator → {Trainer, Front Desk}.
- Design the share rule Trainer_to_Administrator on the gym forms.
- Understand how record ownership differs from record visibility.

2. Theory

User management decides who may open an application and what each person can see or change. In Zoho Creator, users are linked to permission sets and roles; under Settings → User Permissions these controls appear on the Permissions, Roles and Data Sharing tabs. RBAC assigns rights to roles instead of individuals, which keeps administration simple, supports least privilege and is easy to audit.

Concept	Meaning
Users	People added to the application who sign in to work with its forms, reports and data.
Permissions	Named sets that define what a user may do (for example Read and Write).
Roles	Positions in the organisation tree that decide where a user sits and which records are visible.
Role hierarchy	A higher role normally sees the records of the roles below it, not the reverse.
Data sharing	Rules that extend visibility between roles or users at Read Only or Read/Write level.

3. Users of the gym application

User	Role	Intended work
pranavsrinevas.v.2024.csbs@rajalakshmi.edu.in	Administrator (app owner)	Full control: trainers, plans, members, classes, payments, attendance, reports, dashboard, users
sureshbalajiii55577@gmail.com	Trainer	View class schedule and members; mark attendance; view reports

4. Procedure (to be carried out in Settings → User Permissions)

1. Open the gym application in Edit mode and choose the Settings tab.
2. Under Permissions → User Permissions open the Permissions tab and review the Read and Write sets.
3. Open the Roles tab and arrange: CEO at the top, Gym Administrator beneath it, and Trainer and Front Desk beneath the Administrator.
4. Open the Data Sharing tab, choose New Share Rule, name it Trainer_to_Administrator, select the form (for example Attendance_Tracking), share from Roles → Trainer to Roles → Gym Administrator, and set the level of access to Read Only.
5. Save the rule, confirm that it appears in the Data Sharing list and reopen it with Edit Sharing Rule to check every field.

[Screenshot to be inserted]
Insert the screenshots of the Permissions tab, Roles tab, New Share Rule panel and the saved rule once configured.

5. Role hierarchy and data sharing

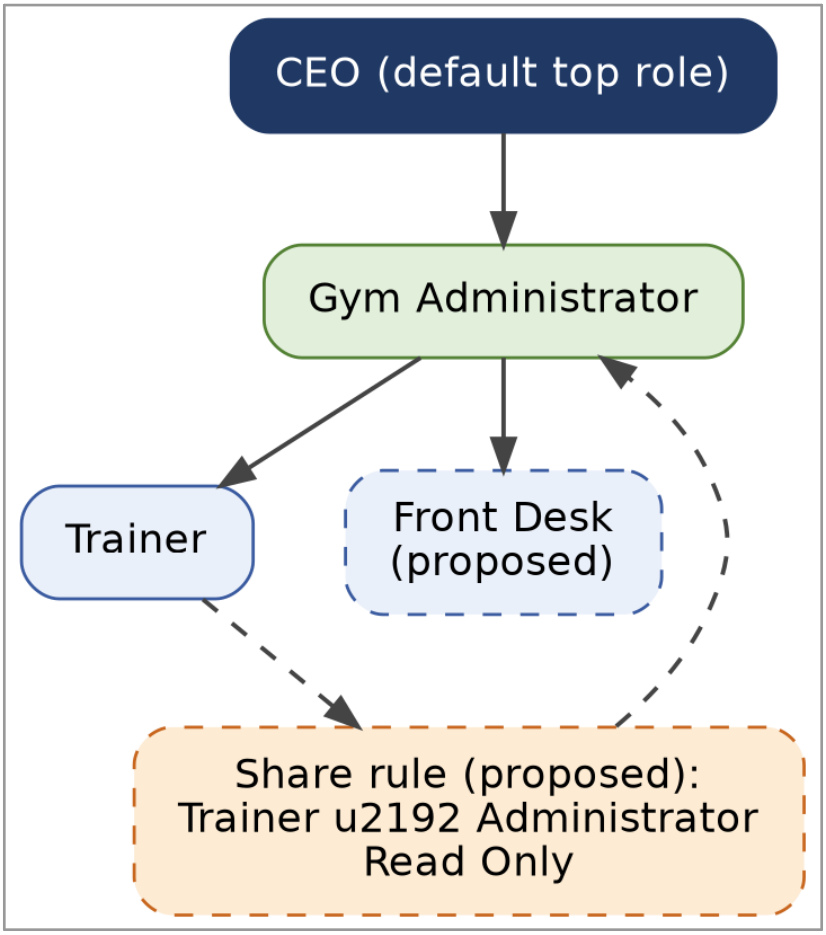


Figure 1: Proposed role hierarchy with the Trainer → Administrator share rule (design).

By default a hierarchy gives upward visibility only. A share rule adds a deliberate, auditable channel; here the Trainer's attendance records become visible to the Administrator without giving the Administrator system-wide permission changes.

Proposed permission matrix

Component	Administrator	Trainer
Trainer_Registration, Membership_Plans	Add, Edit, View, Delete	View
Member_Registration	Add, Edit, View, Delete	View
Class_Schedule	Add, Edit, View, Delete	Add, Edit, View
Payment_Tracking	Add, Edit, View, Delete	No access
Attendance_Tracking	Add, Edit, View, Delete	Add, Edit, View
Reports and Gym Dashboard	View	View

This matrix is a proposed design. The actual permission and role settings should be verified in the live application before the screenshots above are added.

6. Result and Conclusion

The user roles of the gym application were defined (Administrator and Trainer), and a role hierarchy and a Read Only data-sharing rule were designed on the basis of least privilege. The experiment shows how permissions, roles, the hierarchy and data sharing together control access in Zoho Creator.